Large Language Models in Data Science

Week 5: Documents, Retrieval and Agent Loops

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This Week: Answer the Student's Question

"J'ai perdu ma carte étudiante. Comment la remplacer ?"

Last week's task: classify the request

Output: the label `get_student_card`.

This week's task: answer the question directly

Give the LLM the question and the relevant documents. Ask it to explain the replacement procedure and cite its sources.

Documents for Different Student Services

Our collection covers several services. D2 and D3 answer the lost-card question. These are fictional teaching documents, not AMU policies.

ID	Title	Text
D1	Card activation	Activate a new student card through the student portal.
D2	Lost student card	If you lose your student card, request a replacement at the student office.
D3	Collecting a replacement card	Collect the replacement card at the student office and bring photo ID.
D4	Library loans	Renew a library loan through your library account.
D5	Campus Wi-Fi	Connect to the campus Wi-Fi using your university login.
D6	Course timetable	Find lecture times and rooms in the timetable portal.

Start by Supplying All the Documents

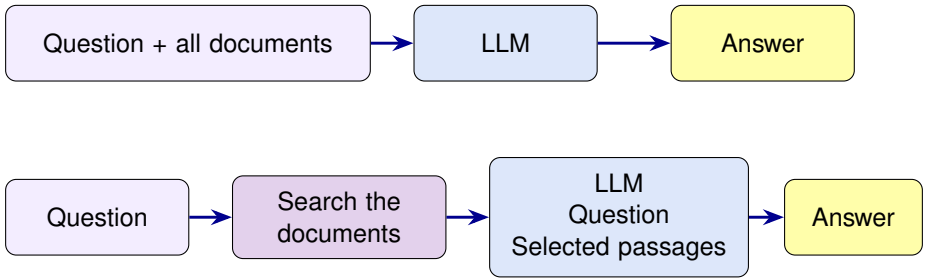
These documents are short enough to send together. The document text supplied with the question is its **context**.

```
System instructions:
Answer in French using the supplied documents.
Cite the document IDs. Say when information is missing.

User message:
DOCUMENTS:
[D1] Activate a new student card through the student portal.
[D2] If you lose your student card, request a replacement at the student office.
[D3] Collect the replacement card at the student office and bring photo ID.
[D4] Renew a library loan through your library account.
[D5] Connect to the campus Wi-Fi using your university login.
[D6] Find lecture times and rooms in the timetable portal.
QUESTION:
J’ai perdu ma carte étudiante. Comment la remplacer ?
```

Retrieve Passages Before Asking the LLM

Retrieval-augmented generation (RAG) searches documents and adds the selected passages to the model's prompt.

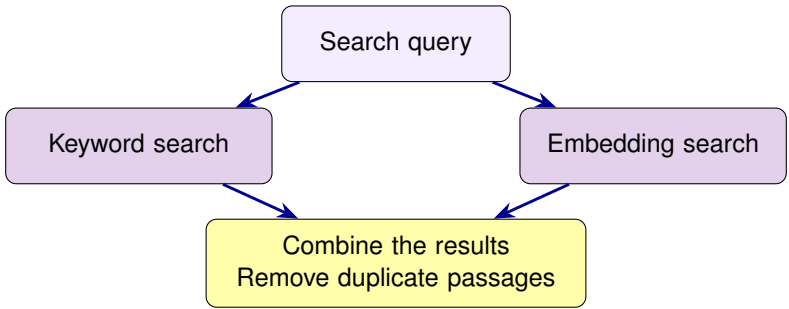


Search by Words or by Meaning

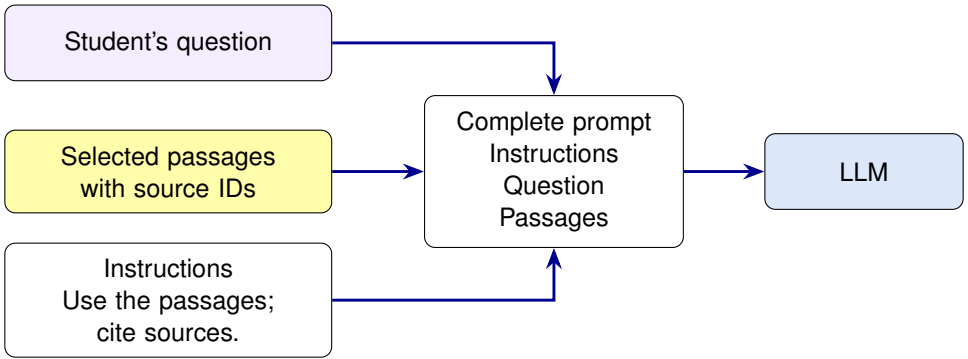
We use the full student question as the search query: the text sent to the search function.

Keyword search matches query words to passage words. BM25 is a scoring method that also accounts for word frequency and passage length.

Embedding search uses the sentence vectors from Week 4. Compare the question's vector with each passage's vector to find similar text.



Put the Selected Passages into the Prompt



Did the Search Find the Information We Need?

Question sent to search

J'ai perdu ma carte étudiante. Comment la remplacer ?

Before evaluating search, we read the documents and identify the relevant passages:

- D2: where to request a replacement.
- D3: where to collect it and what to bring.

We compare the returned documents with this reference set:

$$\text{Precision} = \frac{\text{Relevant passages returned}}{\text{All passages returned}} \qquad \text{Recall} = \frac{\text{Relevant passages returned}}{\text{All relevant passages (2)}}$$

Check the Answer Against Its Sources

Example answer using D2 and D3

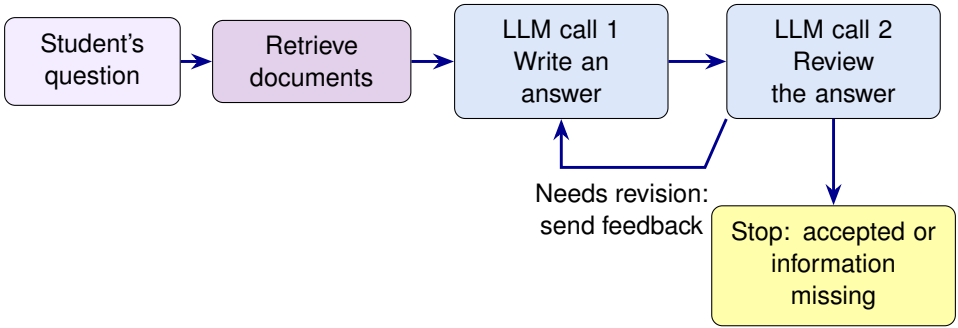
Demandez le remplacement au service de scolarité [D2]. Récupérez la carte auprès de ce service et apportez une pièce d'identité [D3].

Check each instruction against its cited passage.

If the student also asks "Combien coûte le remplacement ?", none of our documents gives a price. The answer should state that the price is missing.

Generate an Answer, Review It, Then Revise

A second LLM call checks whether the answer addresses the question and is supported by the documents. Both calls receive the original question and the retrieved passages.



The Answer–Review Loop in Python

Each attempt makes two LLM calls: `write_answer` drafts or revises the answer; `review_answer` checks it against the question and passages.

```
passages = retrieve(question)
answer, feedback = "", ""
for attempt in range(4):
    answer = write_answer(
        question, passages, answer, feedback)
    review = review_answer(
        question, passages, answer)
    if review["status"] != "revise":
        break
    feedback = review["feedback"]
```

Four attempts mean at most eight LLM calls.

The review returns a status and feedback:

- `accepted`: answer complete and supported.
- `revise`: correct the problems in the feedback.
- `missing_information`: sources insufficient.

After attempt four, report any remaining problems.

Replace the Teaching Documents with AMU Webpages

We now collect real sources for the same student services.

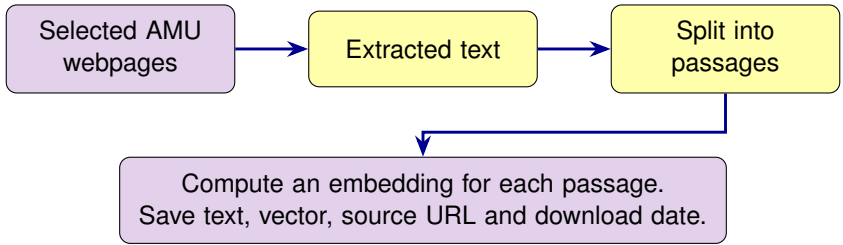
Make the student's situation explicit

J'étudie à l'IUT d'Aix-Marseille. J'ai perdu ma carte étudiante. Comment la remplacer ?

Official source	Information to collect
IUT: services de la scolarité	Card replacement instructions for IUT students.
DirNum: réseau Wi-Fi	Connecting to the university Wi-Fi.
BU: emprunter / réserver	Borrowing, renewing and reserving documents.

Collect the Pages and Make Them Searchable

Use Python `requests` to download each page and `BeautifulSoup` to extract its main text, excluding menus and navigation.



`retrieve(question)` now selects from these collected AMU passages for the answer–review loop.

Our RAG or a Model with Web Search?

Send the same student questions through three workflows:

Workflow	Where the evidence comes from	How it answers
Our RAG	Our saved AMU pages	Retrieve, then answer.
Our RAG + review	The same saved pages	Answer, review and revise.
Model + web search	Pages found during the request	Search online; answer with sources.

Use `gemini-3.5-flash-lite`: RAG calls use the free API tier; the web comparison uses Google Search in the free AI Studio interface.

Compare Answers on the Same Student Questions

Prepare questions about card replacement, Wi-Fi and library services. Read the official pages and record the expected facts and supporting URLs before running the comparison.

- **Answer quality:** are the instructions correct and complete for the student's situation?
- **Sources:** do the cited pages support the claims and apply to the right faculty or service?
- **Missing information:** does the system explain what it cannot establish?
- **Time and tokens:** count every notebook answer and review call. Record browser timings separately.

Lab: Build and Evaluate an AMU Assistant

1. Download the selected AMU pages and inspect the extracted text.
2. Split the text into passages; keep source URLs and compute embeddings.
3. Answer student questions with RAG, then add the answer–review loop.
4. Ask the same questions with Google Search enabled in free AI Studio.
5. Check the answers against the sources and compare quality, time and cost.

Further Reading

- Lewis et al.: Retrieval-Augmented Generation — the original paper.
- Google: long context — supplying documents directly to a model.
- Google: web search with Gemini — search integration and source citations.
- Anthropic: building effective agents — fixed workflows and model-selected actions.
- Hugging Face Agents Course, Unit 1 — tools, actions and observations.